

Question 4

Not yet answered

Marked out of
5.00

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Use the following code as a component in your top level design to build a **3-bit Subtractor** and compile it in Quartus II and upload a print screen showing the successful compilation with the code. **The top level design entity name should be your student ID:**

entity **FA** is

port (A,B,Cin: in std_logic;

S, Cout: out std_logic);

end **FA**;

Architecture arch of **FA** is

begin

Cout <= Cout <= ((A xor B) and Cin) or (A and B);

S <= (A xor B) xor Cin;

end arch;

☰ Quiz navigation

1	2	3	4	5
7	8	9		

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Question 4

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The truth table below represents a FSM with a 1-bit input (x) and a 1-bit output y . Suppose the machine is initialized to its starting state and then processes the input values: 1,0,0,1,0,0 (1 input bit after each Clk). What are the final state?

Present State		Next State				Output	
		$x = 0$		$x = 1$		$x = 0$	$x = 1$
A	B	A	B	A	B	y	y
0	0	0	0	0	1	0	0
0	1	0	0	1	1	1	0
1	0	0	0	1	0	1	0
1	1	0	0	1	0	1	0

Select one:

- ☐ a. 01
- ☐ b. 00
- ☐ c. 10
- ☐ d. 11

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PREVIOUS ACTIVITY

Report submission Lab 7 Part II

Question 5

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To design a 0 to 8 counter we will need how many D-flip flops?

Select one:

- ☐ a. 3
- ☐ b. 5
- ☐ c. 4
- ☐ d. 6

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PREVIOUS ACTIVITY

Report submission Lab 7 Part II

Question 5

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To design a 9 to 0 counter we will need how many D-flip flops?

Select one:

- ☐ a. 5
- ☐ b. 4
- ☐ c. 6
- ☐ d. 3

☰ Quiz navigation



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PREVIOUS ACTIVITY
Report submission Lab 7 Part II

Question 3

Not yet answered

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Say that $a(7:0) = "00110011"$ and $b(3:0) = "1111"$. Determine the values produced by the assignments below:

$a(7 \text{ DOWNTO } 4) \text{ XOR NOT } b;$

Answer:

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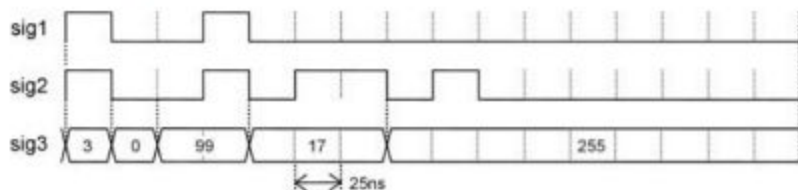
```

signal X : std_logic_vector(7 downto 0);
signal Y : integer;
signal Z : unsigned(7 downto 0);

```

- $Y \leftarrow \text{to_integer}(X);$
- $Y \leftarrow \text{to_integer}(\text{unsigned}(X));$
- $X \leftarrow \text{std_logic_vector}(Z);$
- $Z \leftarrow \text{std_logic_vector}(X);$
- $X \leftarrow \text{to_integer}(\text{unsigned}(Y));$
- $X \leftarrow \text{to_integer}(Y);$
- $X \leftarrow \text{unsigned}(Z);$
- $Z \leftarrow \text{to_integer}(X);$

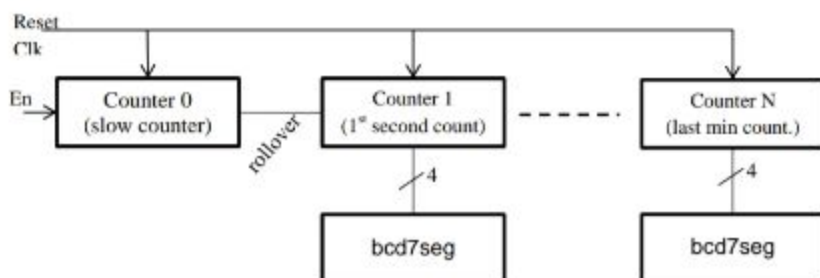
Q2: Write the testbench code to generate the signals in the figure below: (12 marks)



Q3: Assume you have two components which are: (12 Mark)

- "modulo_counter_er": This is a parametrized module counter that we can change its size (n) and the fact that it counts from 0 to (k-1) which are defined as Generic and can be altered in the top level. In addition the output include rollover signal to enable the next counter.
- "bcd7seg": this a BCB (0-9) 7-segment interface to display the output on ports HEX(3-0)

Design and implement a circuit using VHDL code that acts as a real-time clock. It should display the minutes (from 0 to 59) on HEX3 - 2, the seconds (from 0 to 59) on HEX1 - 0, and hours (from 0 to 12) on HEX1 - 0. Stop the clock whenever KEY0 is being pressed and continue the clock when KEY0 is released. Use the board 50 MHz clock in your design. Hint: the figure below hint how the final design should be:



Q4: Consider the state diagram in Figure 4 (15 Mark)

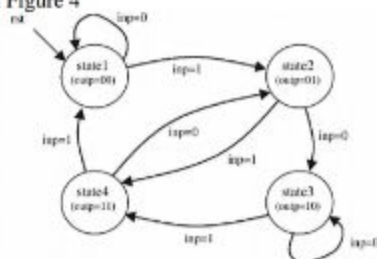
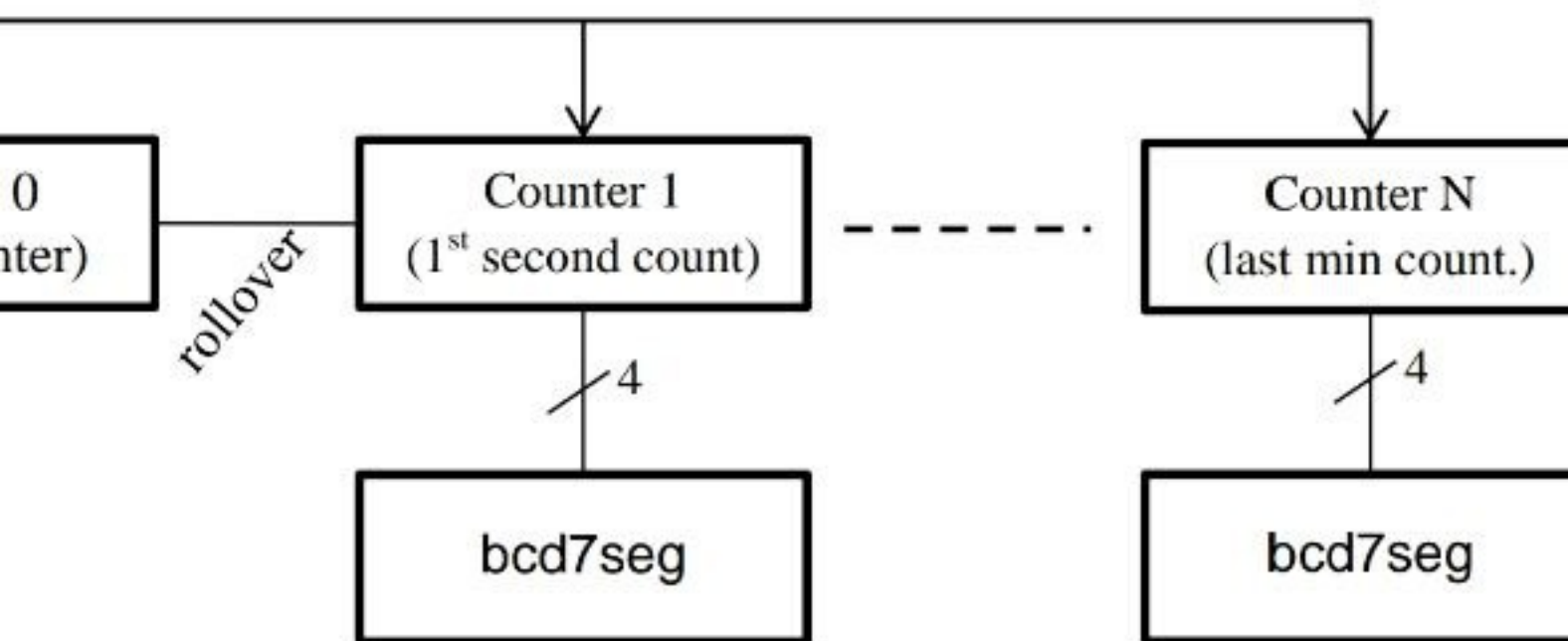


Fig. 4

- Draw the algorithmic state machine (ASM) of this state machine. And how many Flip flop you would need for the implementation?
- Write the VHDL code of state transition rules and output generation for the state machines in the following template



State diagram in Figure 4

(1)

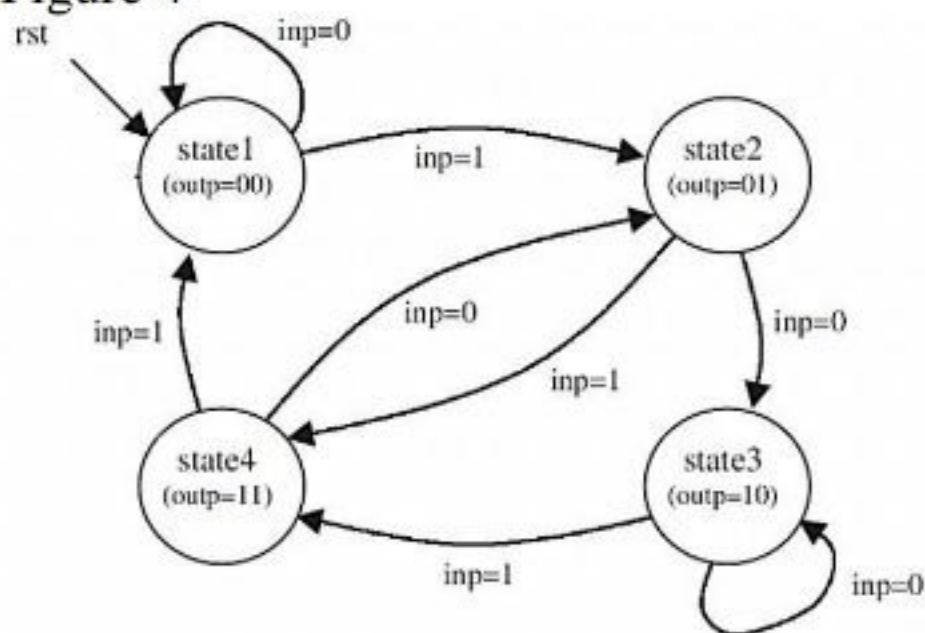


Fig. 4

Algorithmic state machine (**ASM**) of this state machine. And how many FLOPs are required for its implementation?

Write the Verilog HDL code of state transition rules and output generation for the state machine. Use the template

```

LIBRARY ieee;
USE ieee.std_logic_1164.ALL;

ENTITY fsm IS
PORT (
    x, time1, time2, clk, rst: IN std_logic;
    y: OUT std_logic);
END ENTITY;

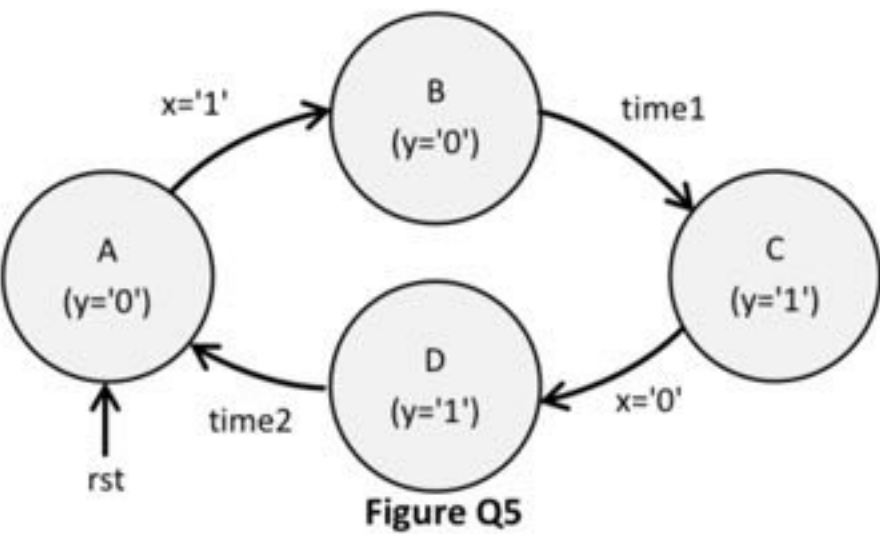
ARCHITECTURE fsm OF fsm IS
    TYPE state IS (A, B, C, D);
    SIGNAL present_state, next_state: state;
BEGIN
    PROCESS(clk)
    BEGIN
        IF(rising_edge(clk)) THEN
            IF(rst = '1') THEN
                present_state <= A;
            ELSE
                present_state <= next_state;
            END IF;
        END IF;
    END PROCESS;

    PROCESS(x, time1, time2, present_state)
    BEGIN
        CASE present_state IS
            WHEN A =>
                IF x = '1' THEN
                    next_state <= B;
                ELSE
                    next_state <= A;
                END IF;
            WHEN B =>
                IF time1 = '1' THEN
                    next_state <= C;
                ELSE
                    next_state <= B;
                END IF;
            WHEN C =>
                IF x = '0' THEN
                    next_state <= D;
                ELSE
                    next_state <= C;
                END IF;
            WHEN D =>
                IF time2 = '1' THEN
                    next_state <= A;
                ELSE
                    next_state <= D;
                END IF;
        END CASE;
    END PROCESS;

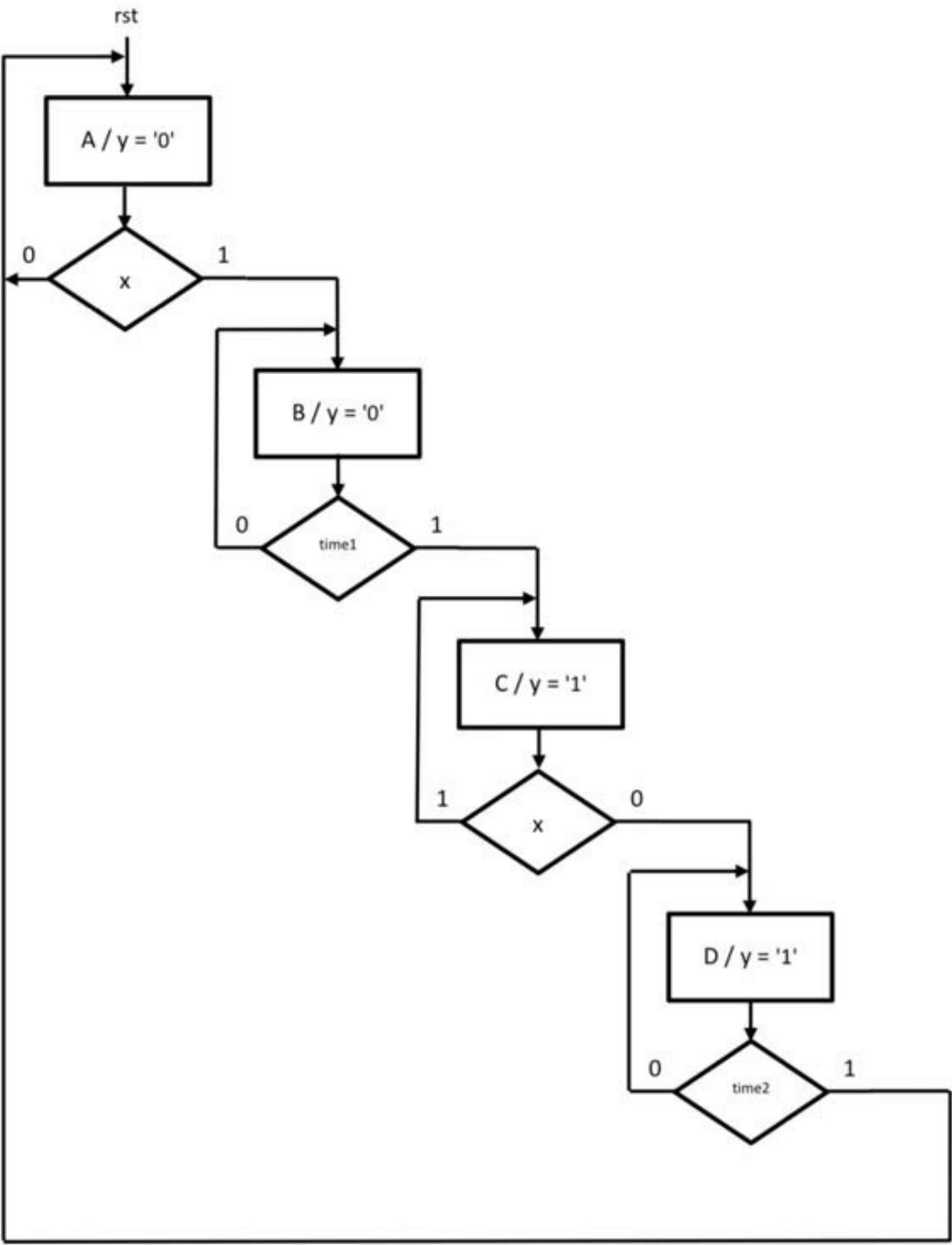
    y <= '0' WHEN present_state = A OR present_state = B ELSE
        '1';
END ARCHITECTURE;

```

Q5- Consider the state diagram in Figure Q5



(A) Draw the algorithmic state machine (ASM) of this state machine. (5 Marks)





Calculators are not allowed

Note: Answer all questions

Q1: Say that $a(7:0) = "00110011"$ and $b(3:0) = "1111"$. Determine the values produced by the assignments below. (answer only 5) (5 marks)

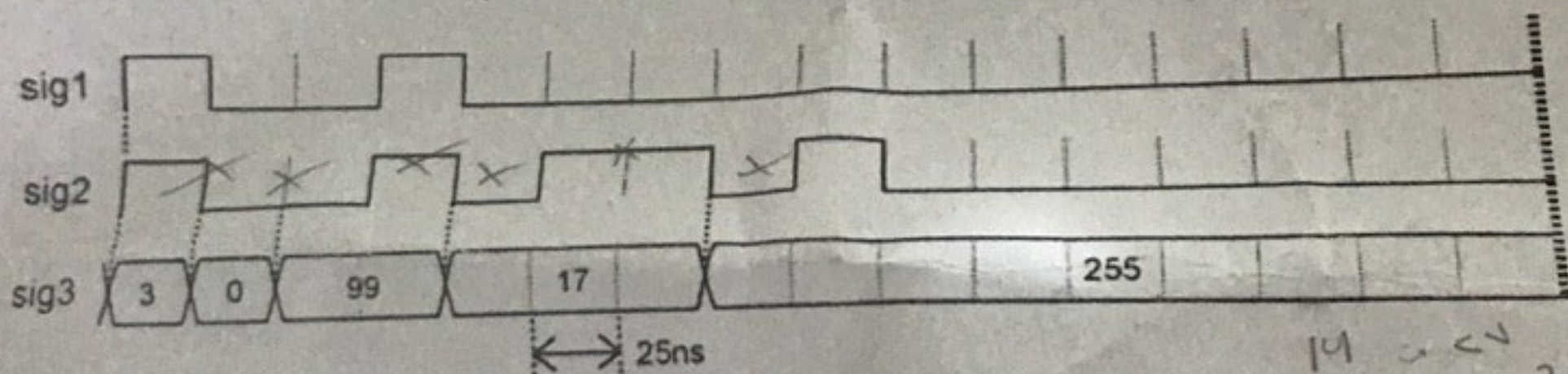
- $a(7 \text{ DOWNTO } 4) \text{ NAND } "0111"$
- $a(7 \text{ DOWNTO } 4) \text{ XOR NOT } b$
- $"1111" \text{ NOR } b$
- $b(2 \text{ DOWNTO } 0) \text{ XNOR } "101"$
- $a(2 \text{ DOWNTO } 0) \& "000"$
- $a(5) \& b(5) \& a(5 \text{ DOWNTO } 2)$

5432 1100

Q2: Implement a 4x1 multiplexer with: (10 marks)

- where inputs are single bit and using only logical operators (gate level design).
- N-bit inputs specify N using GENERIC and using WHEN statement.

Q3: Write the testbench code to generate the signals in the figure bellow: (10 marks)



Q4: Write the code for a zero to nine counter with reset signal input and the output (state value) is displayed using 7-segment displays HEX0. (15 mark)

Dr. Ammar Jallawi Mahmood

Good luck although you shouldn't need it!